* What is electrolysis? State and explain the Kohlrausch's law of independent migration of ions.

* Describe the Calomel reference electrode and give half-cell reaction when it is coupled with a standard hydrogen electrode.

* Difference between an electrolytic cell and galvanic cell. Write the cell reactions for the following electrochemical cells:
(i) $Zn/Zn^{2+}\,||\,Cu^{2+}/Cu$; (ii) $Fe/Fe^{2+}\,||\,Ag^{+}/Ag$;

* Define the following terms:
(i) Specific conductance (ii) Electromotive force
(iii) Liquid junction potential

* State the Faraday's law of electrolysis. Mention its importance.

* 0.1978g of copper is deposited by a current of 0.5 ampere in 45 minutes. What is the electrochemical equivalent of copper?

* What is a salt bridge? Describe the function of a salt bridge.

* Discuss charge and discharge chemistry of Lead-acid storage battery.

* Calculate the EMF of the following cell:
$Zn/Zn^{2+}(0.002M)\,||\,Cu^{2+}(0.01M)/Cu$
Given that $E^{\circ}_{Cu^{2+}/Cu} = 0.34V$ and $E^{\circ}_{Zn^{2+}/Zn} = -0.76V$.

Electrochemical cell reactions

1) Anode reaction : $Zn \longrightarrow Zn^{2+} + 2e$

Cathode reaction : $Cu^{2+} + 2e \longrightarrow Cu$

---

cell reaction: $Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$

II) Anode reaction : $Fe \longrightarrow Fe^{2+} + 2e$

Cathode reaction : $2Ag^{+} + 2e \longrightarrow 2Ag$

Cell reaction : $Fe + 2Ag^{+} \longrightarrow Fe^{2+} + 2Ag$

$t = 45 \text{ min} = 45 \times 60 s = 2700 s$

$I = 0.5 A =$

$Q = It = 0.5 \times 2700 = 1350 C.$

Amount of copper deposited by 1 coulomb $= \frac{0.1978}{1350}$

$= 0.000146 g$

* What are electrochemical series? Give its application.
* Discuss Arrhenius theory of electrolytic dissociation.
* Zn gives H_2 gas when it reacts with H_2SO_4 but Ag does not. Explain (given that $E^0_{Zn^{2+}/Zn} = -0.76V$ and $E^0_{Ag^+/Ag} = 0.80V$).
* Derive an expression for the radius of nth orbit of H-atom.
* State and explain the Pauli exclusion principle.
* Find out the four quantum of 15th electron of K.
* Find out the frequency in terms of wave number($\bar{\nu}$) for the 2nd spectral line of Lyman series. Given that $R_H = 1.097 \times 10^7\ m^{-1}$.
* Classify the elements on the basis of the electronic configuration of their atoms.
* Define ionization energy. The first ionization energy of nitrogen is higher than that of oxygen- explain.
* Define covalent radius. Explain why the atomic radius of the elements decreases from left to right in a period whereas it increases on moving down in a group.
* What is 'octet rule'? Draw the Lewis (on electron dot) structure of the following molecules/ions:
  i) $SiCl_4$ ii) NH_4^+ iii) NO iv) PCl_5

Actinides: In the atoms of 5f-block elements the extra electron enter 5f-orbitals. in the atoms of the elements of this block 5f orbitals are being progressively filled. A group of 14 elements from thorium, Th (Z=90) to lawrencium, ~~Lr~~ lw (Z=103) Constitute this series. The elements of this series are called actinides.

which of the have maximum magentic moment

Mn^{2+}, Cr^{2+}, Fe^{2+}

Ans: $Mn^{2+}(25) \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5 \rightarrow [Ar]\, 3d^5$

$Cr^{2+}(24) \rightarrow [Ar]\, 3d^4$

$Fe^{2+}(26) \rightarrow [Ar]\, 3d^6$

Mn^{2+} has maximum nob of unpaired electrons = 5 wheas Cr^{2+} has 4 and Fe^{2+} 4 So

magnetic moment i's $= \sqrt{n(n+2)}$ · B·M

where n i's the no of unpaired electrons, So the Mn^{2+} has maximum number of unpaired electrons. of m.m

* Explain why H_2O is liquid whereas H_2S is gas at NTP.
* Define hydrogen bond. Classify the hydrogen bond with examples
* What is covalent bond? Illustrate the formation of covalent bond.
* What is metallic bond? Deduce the shape of PCl_5 and SF_6 molecules with the help of hybridization concept.
* What are actinides? Write down some of their uses.
* Which one of the following has the maximum magnetic moment: Mn^{2+}, Cr^{2+}, Fe^{2+}
* Write down some general properties of p-block elements.

~~Sol~~ These elements are located at the extreme right of the periodic table.

i) They are metals, nonmetals, metalloids, inert gases
ii) electronic configuration as ns^2, np^{1-6}
iii) High melting and boiling point
iv) good conductor of electricity
v) Component of glass industry
vi) metalloids uses for microchips
vii) Nonmetals are importants for living things